



Product Innovation

! Action-to-Button Mapping !

Sensors

accelerometer, altimeter, gyroscope
magnetometer, pressure-sensitive grid

Controller Input Mapping Example

Physical Action	Technical Parameters	Controller Response
High-Frequency Stepping	Step rate ≥ 120 steps/min (Speed maps to 0-100% thrust)	Left Joystick Forward or Back Push
Waist Left-Right Rotation	Gyroscope detects rotation angle $\pm 30^\circ$ (Error $< 1.5^\circ$) ($30^\circ/\text{s} = 90^\circ$ in-game turn)	Left Joystick Left or Right Push
Grip Squeeze	Pressure sensor dynamic detection (5-15kg) (Silicone layer deformation feedback) (5kg=50% travel/15kg=100% charge)	RT Trigger
Arm Raise & Hold	EMG sensor sustained activation $> 3\text{s}$ (Posterior deltoid activation) (3s activation/5s enhancement)	Y Button Long Press
Pushup Pulse	Pectoral EMG signal + Height sensor (Ground reset detection) (1 standard rep = 1 button press)	X Button Cycle Trigger